

Evaluating the Value of Topology-Aware Features for Cyber-Physical Grid Monitoring: Detection, Localization, and Real-Time Feasibility

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Abstract—Distinguishing a cyber false-data-injection attack (FDIA) from a legitimate physical disturbance, localizing it, and doing so within a protection-relevant time budget are usually studied separately. We study all three, across three networks (a synthetic inverter-dominated 5-bus microgrid and two standard IEEE test systems, 14-bus and 30-bus) with a weighted-least-squares (WLS) AC state estimator, built specifically to test whether explicit topology-relational features (neighbor- and incident-edge-comparison statistics) help. A first pass on the 5-bus testbed appeared to show they did: `topology_fusion` beat a prior-based baseline at both detection (balanced accuracy 0.950 vs. 0.873) and localization (top-1 accuracy 0.980 vs. 0.780), confirmed across four independent seeds. That comparison was not a fair one. Topology-relational columns are named after the residual/innovation quantities they compare (e.g. `neighbor_mean_residual`), and a name-substring-based feature-set split let roughly half of them leak, mislabeled, into what were meant to be topology-free baselines – so `topology_fusion` was effectively being compared against baselines that already contained most of its own information. Correcting this and building a genuinely topology-free ablation – residual and prior-innovation features combined, with no relational comparisons at all – matches `topology_fusion` on detection (gap -0.003 ± 0.010 across 4 seeds, sign unstable, the signature of noise around zero) and is narrowly, consistently ahead of it on localization (gap -0.003 ± 0.007 , never favoring topology but only 0.3 points, an order of magnitude smaller than the original uncorrected gap this ablation was built to explain). We report this as the headline **5-bus finding**: combining residual and prior-innovation information explains the entire measured advantage over `prior_only` alone; explicit topological relational structure adds nothing further, and this fusion gain is itself real and large, not a topology effect. A separately-implemented, size-agnostic pipeline on IEEE 14-bus needed an analogous correction – an independent rediscovery of the same category of mistake. Once fixed and confirmed across four further independent seeds, detection again shows no topology advantage (gap -0.015 ± 0.004 , `topology_fusion` trailing every seed; unlike at 5-bus, fusion itself adds nothing over residual-only here either, both at 0.750); localization reverses: `topology_fusion` keeps a small, seed-robust lead (gap $+0.018 \pm 0.010$, never flipping) – a genuine, replicated advantage, not a second null result. A third network (IEEE 30-bus), tested the same way after an initial single run appeared to reproduce this reversal, does not confirm it: both gaps are small and change sign across seeds (detection $+0.001 \pm 0.009$, localization $+0.005 \pm 0.018$) – the single run had simply landed on a representative-looking seed by chance. Across all three networks,

topology-relational features show no reproducible detection benefit; for localization, the benefit is not universal either, appearing reproducibly only at IEEE 14-bus. We separately profile the full decision pipeline against a one-cycle (20 ms at 50 Hz) protection-relevant computational target and find the original 43 ms pipeline slow because of an avoidable software inefficiency, not the model or the state estimator’s iteration count; fixing it brings the pipeline to 16.4/17.0/17.1 ms at the median/p95/p99. A third, independent result further scopes any deployment claim: the detector does not generalize to a completely unseen attack type (0–4% recall when one of two attack families is withheld). We think catching and correcting our own feature-labeling bug, rather than reporting the more publication-friendly original result, is the more useful contribution here.

Index Terms—false data injection attack, state estimation, cyber-physical systems, attack localization, real-time systems, feature ablation, inverter-based resources

I. INTRODUCTION

Modernizing power grids toward higher penetrations of inverter-based resources (PV, battery storage, grid-forming converters) changes the character of protection and monitoring in two ways relevant here. First, inverters provide far less fault current than synchronous generators, which strains protection schemes designed around synchronous-machine fault signatures [1]. Second, the same measurement infrastructure used for state awareness is now also an attack surface: an adversary who can spoof a subset of sensor readings can attempt a false-data-injection attack (FDIA) against the state estimator that is, by construction, statistically indistinguishable from a legitimate measurement under a residual-based bad-data test [2].

Renewable-heavy grid infrastructure specifically is not a hypothetical target: in December 2025, a destructive cyber-attack against Poland’s energy sector – corrupted firmware forced into endless restart cycles across roughly 30 wind and solar farms and a heat-and-power plant, during a cold-weather period, attributed to infrastructure overlapping with the Dragonfly/Static Tundra threat actor – was contained before it caused a blackout affecting an estimated 500,000 people [3]. That incident was an OT/firmware-integrity attack rather than a state-estimation FDIA specifically, but it establishes

the same operational stakes this paper’s threat model targets: an operator or automated scheme facing a live control-plane anomaly, under real time pressure, in exactly this class of infrastructure.

A control-room operator, or an automated protection scheme, therefore faces a three-part problem for any anomalous reading: *is this a cyberattack or a physical event, where is it*, and – if it is to inform a protective action rather than only a post-hoc forensic one – *can a decision be reached inside a protection-relevant time budget*. These three questions are usually studied separately. A large FDIA-detection literature (Section II) asks the first question, typically evaluated offline against a fixed dataset. A smaller attack-localization literature asks the second. Real-time feasibility is addressed by a much smaller, very recent body of work – notably a 2026 benchmark [4] that found deep classifiers for this exact joint task are individually fast (sub-15 ms) but the measured end-to-end pipeline is not (50–90 ms), calling this “a critical gap between algorithmic capability and protection-grade deployment” and recommending hardware acceleration. A separate, popular line of work asks whether giving a detector explicit *topology* – not just raw sensor residuals, but how each meter’s reading compares to its electrical neighbors – improves on any of this; that question is this paper’s own starting point, and its central subject.

This paper studies all three questions on one testbed, and – in the process of doing so honestly – surfaces three corrections to its own early results that we think are worth reporting as part of the method, not edited out of the narrative:

- 1) **A near-perfect early result was wrong, and wrong for a boring reason.** An initial evaluation of topology-aware detection reported 1.000 balanced accuracy across every metric and three different model families, on a 72-sample test set. We show this was not a data-leakage bug (we checked, fixed a real but inconsequential defensive gap, and the result did not change) but simply that the attack scenarios were too easy. Hardening the attack magnitude toward the sensor noise floor initially *reversed* the finding at small sample size ($n=72$) before a larger run ($n=300$) showed the original small-sample reversal was itself noise.
- 2) **The resulting “topology helps” finding did not survive its own ablation – twice.** `topology_fusion` was compared against `residual_only` and `prior_only`, but not against a set that combines the two *without* any relational features – the actual, fair control. Building that control exposed a feature-labeling bug at 5-bus: explicit topology-relational columns are named after the residual/innovation quantities they compare (e.g. `neighbor_mean_residual`), so a name-substring-based split had silently let about half of them leak into what were meant to be topology-free baselines. Auditing the separately-implemented IEEE 14-bus pipeline (Section VII) the same way found an analogous flaw there too – an explicit list inclusion rather than a substring collision, but the same problem.

Once both are corrected, the genuinely topology-free `residual_plus_prior` ablation matches or slightly beats `topology_fusion` for detection at every scale tested; for localization, the ablation is narrowly ahead at 5-bus but `topology_fusion` keeps a small, 4-seed-robust edge at IEEE-14 – the one place in this paper topology shows a real, if modest, advantage, one that a third network (IEEE 30-bus) does not reproduce (Section IV, Section VII).

- 3) **The pipeline-latency gap reported in concurrent work is, at least here, mostly fixable software engineering, not an algorithmic ceiling.** We independently reproduce the qualitative finding of [4] (fast model, slow pipeline) and then localize the actual bottleneck: not the classifier, not the state estimator’s Newton-type iteration count, but a measurement-table object being rebuilt from scratch through a general-purpose element-creation API every cycle, when its structure never changes. Fixing that one thing accounts for most of the gap closed.

Contributions:

- A methodologically rigorous, network-specific result: once an (independently rediscovered, twice) feature-labeling bug is corrected and a fair, same-information ablation is built, explicit topology-relational features measure no reproducible detection benefit over combining residual and prior-innovation features, at any of three networks tested (5-bus, IEEE 14-bus, IEEE 30-bus), each confirmed across four independent seeds (5-bus: also across two stress conditions), using two unrelated feature-engineering implementations that each independently needed the identical fix; for localization, the same corrected comparison instead reveals a genuine, seed-confirmed advantage that is not universal either – reproducible only at IEEE 14-bus, with no robust advantage at 5-bus or IEEE 30-bus (Section IV, Section VII) – together with a structural, quantitative explanation for the 5-bus null result, and a direct test that rules the same explanation out at IEEE-14 (Section IV, Section VII).
- A root-cause profiling methodology for the “fast model, slow pipeline” gap identified in prior work, and a demonstrated fix that closes most of it without hardware acceleration (Section V).
- A further explicit negative result reported as a scope condition rather than omitted: no zero-day attack-type generalization (Section VI).

II. RELATED WORK

Foundational FDIA and stealth constructions. The FDIA threat model used throughout this paper – an attacker with knowledge of the network model constructing a measurement perturbation that leaves the WLS residual unchanged – follows Liu, Ning & Reiter’s foundational construction [2] and its AC-consistent extensions [5]. A growing subline constructs such attacks with *less* attacker knowledge than the original construction assumes: without line parameters [6], or fully “blind” to topology using generative or physics-informed models [7], [8]

– all harder threat models than the one used here and natural next stress conditions for this project’s own attack generator. In the opposite direction, low-magnitude attacks close to the sensor noise floor – the specific regime Section IV’s hardened attack magnitude targets – have their own dedicated detection literature [9], worth comparing against directly in future work.

Deep-learning FDIA detection. A substantial literature applies CNN/LSTM/transfer-learning architectures to FDIA detection [10]–[12], generally evaluated as a binary classification task on a fixed attack distribution. This project’s *residual_only* and *prior_only* baselines sit in this tradition; its contribution is isolating what *explicit topology-relational* features specifically add beyond a fair combination of the two, under a stress-tested (not default-easy) attack distribution.

Topology-aware / graph-based state estimation and detection. The closest published framing to this paper’s central question is topology-aware graph neural network (GNN) based state estimation [13]–[16], which use graph convolution/attention to exploit network structure. Checking the closest of these directly – full text, not abstracts, specifically to avoid the kind of overclaim this paper has already had to correct twice in itself – shows the picture is mixed, not uniformly missing the check this paper’s Section IV performs: [13] and an earlier, closely related study by an overlapping author group [14] both compare their GNN against a same-feature “regular” DNN baseline (identical PMU-derived node features, no adjacency matrix) – a genuine, correctly-scoped topology ablation, of exactly the kind this paper’s own original comparison lacked; [15]’s non-graph baseline, by contrast, is not clearly confirmed on the text available to us to receive identical input features. We therefore do *not* claim this specific ablation is generally absent from the GNN-for-power-systems literature – checked directly, in at least two cases it is not. This paper’s contribution is narrower, and we think still useful: a same-architecture, same-feature comparison (GNN vs. DNN, as in [13], [14]) checks whether a different *model* needs topology; it does not by itself catch a bug living one level lower, in how the “non-topological” feature *set* is defined in code, before any model sees it. Section III’s column-name collision produced exactly that kind of baseline contamination – undetectable by an architecture-level ablation alone, since the corrupted *residual_only/prior_only* sets were never fed to a genuinely different, topology-blind model in the first, uncorrected comparison at all. Section IV’s out-of-fold linear-redundancy check is offered as a second, complementary, and considerably cheaper pre-check for this specific failure mode: verifiable on the feature matrix alone, before training any model, topology-aware or not.

Attack/event localization. Localization work spans capsule networks for load-altering (not FDIA) attacks evaluated across 14/39/57-bus systems [17], moving-target-defense-assisted localization [18], and online detection-plus-localization schemes [19]. None of the abstracts available to us report a localization method’s sensitivity to network topology/meshedness directly, or an ablation of relational features against a same-information

non-relational baseline, which is exactly the comparison this paper’s Sections IV and VII run.

Real-time / latency-aware classification. [4] is the direct comparison point for Section V of this paper – it independently measures the same qualitative gap (fast model, slow pipeline) on a different, larger testbed using an industrial electromagnetic-transient (EMT) simulator, and calls for hardware acceleration where this paper instead finds and fixes a software-level cause for a comparable fraction of the gap.

Zero-day / unseen-attack generalization. The general (non-power-specific) intrusion-detection literature widely reports that supervised detectors trained on known attack families do not reliably generalize to unseen ones; a verified, quantitative comparison figure specific to the power-systems FDIA literature is, at the time of writing, still outstanding work (an earlier draft’s figure here did not survive re-verification against its cited source and was removed rather than repeated).

Field-wide self-assessment of adjacent gaps. A 2025 scoping review of ML applications in power-system protection [20] independently reports, at the level of the field rather than any single paper, two gaps this paper runs into directly: most reviewed studies do not discuss computational latency/real-time constraints at all (Section V), and cross-topology generalization is essentially untested field-wide – reviewed work concentrates on fixed IEEE 14/30/39-bus evaluations with no cross-topology testing (Section VII). This positions those sections as addressing gaps the field has already flagged in itself, not problems specific to this project’s own testbed.

III. TESTBED AND METHODOLOGY

Network. A 5-bus, 0.4kV synthetic microgrid built in pandapower: point of common coupling, a PV bus modeled as a reduced-order grid-following (GFL) inverter, a BESS bus modeled as a reduced-order grid-forming (GFM) inverter, and two load buses, connected by 4 lines. `net.f_hz = 50.0` (verified from the model object, not assumed – this fixes the one-cycle protection-relevant computational target below to 20 ms).

State estimation. Weighted least squares (WLS) AC state estimation via `pandapower.estimation.estimate`, with 23 measurements per cycle: voltage magnitude, and P/Q injection, at all 5 buses, plus P/Q from-side flow on all 4 lines – the same model-driven/data-driven/hybrid-driven taxonomy of state estimation under measurement anomalies surveyed in [21] implicitly underlies this paper’s own *residual_only/prior_only/topology_fusion* split.

Threat model. Two synthetic attack families, both targeting a randomly chosen load bus per scenario:

- *Naive*: a single voltage-magnitude sensor is spoofed by a fixed-range random offset.
- *Model-consistent (“stealth”) FDIA*: following the Liu–Ning–Reiter construction adapted to the AC case, the true state is perturbed at the target bus and the corresponding measurement-space attack vector $a = h(x_{\text{attack}}) - h(x_{\text{true}})$ is computed exactly, so the attack is, by construction, invisible to the chi-square residual test.

A benign hard-negative case (physical_load_disturbance, a 25–70% load step at a randomly chosen bus) is included specifically to be confusable with an attack in raw residual terms.

The prior_only feature set, and exactly how its prior is constructed (the leakage-critical detail). Every replication starts from one randomly sampled base operating point (a load/PV level, shared unmodified across that replication’s 4 scenarios). Before any attack or physical disturbance is applied to build the “true” network for a given scenario, an independent copy of that same base operating point has its load values perturbed by a fresh, independently drawn forecast error ($\mathcal{N}(0, 2.5\%)$ per load, clipped to $\pm 6\%$ in the headline configuration) and is solved through its own, separate AC power flow – this perturbed-and-solved copy is the “prior.” prior_only features are the deviation between the WLS state estimate (computed from the scenario’s actual, possibly-attacked measurements) and this prior’s own voltage/angle solution. Critically, the prior is constructed *before*, and entirely independently of, whichever scenario (clean, naive attack, stealth attack, physical disturbance) that replication’s true network becomes – it never sees the true or attacked state, only an independently-perturbed forecast of the same nominal base point. This is what lets stealth attacks, invisible in residual space by construction, become visible at all: the attack is only tuned to be consistent with the WLS residual test, not with an estimate the defender never let it see.

Features. Four feature sets are compared. residual_only: conventional WLS measurement residuals. prior_only: the prior-deviation features defined above. residual_plus_prior: the two combined, with no relational information at all – this is the genuinely topology-free ablation, and the reason it exists is explained next. topology_fusion: residual_plus_prior plus explicit relational features describing each bus relative to its network neighbors (local-minus-neighbor innovation/residual, incident-edge aggregates).

A feature-labeling bug, and why residual_plus_prior exists. This project’s first pass at these results compared topology_fusion only against residual_only and prior_only individually – never against the union of the two, which is the actual topology-free control a skeptical reviewer would ask for. Building that control surfaced a real bug: residual_only/prior_only were originally defined by matching column-name substrings (“residual”, “innovation”), but the explicit relational columns are themselves named after the quantities they compare across neighbors – e.g. neighbor_mean_residual, incident_edge_max_innovation, local_minus_neighbor_residual – so the same substring match silently pulled a large share of every relational column into what were meant to be topology-free baselines (on the 5-bus network: 30 of 65 relational columns into residual_only, 35 of 65 into prior_only). The practical effect was total: residual_only $\cup$ prior_only, as originally defined, equaled topology_fusion exactly, column for

column – so topology_fusion could only ever have appeared to beat each baseline *individually* (a comparison of feature *quantity*, more columns vs. fewer), never a comparison that isolated topology-awareness itself, because no topology-free condition using *all* the same non-relational information had ever actually been run. The fix, applied throughout this paper: exclude columns matching explicit relational-name markers (neighbor_, incident_edge_, local_minus_, local_over_) from residual_only/prior_only, and add their corrected union as the fourth, genuinely topology-free residual_plus_prior set. Section IV reports the corrected comparison against the original.

Models. LogisticRegression, RandomForest, and HistGradientBoosting classifiers are compared against 1-feature interpretable-score baselines, with replication-group-safe 70/15/15 train/validation/test splits and group-aware bootstrap confidence intervals.

Sample-size terminology, defined once here to avoid ambiguity below. A *replication* is one independently sampled base operating point (load level, forecast draw); each replication produces exactly 4 *scenarios* – one clean, one naive-attack, one stealth-attack, one physical-disturbance – so N replications yield $4N$ scenarios in total. The 70/15/15 split is applied at the replication level (never at the scenario level, to keep the 4 matched scenarios from one replication from leaking across train/test), so the resulting *test-scenario count* is $0.15N \times 4$, and the *cyber test-scenario count* used for localization (only the 2 attack scenario types have a real target bus) is half of that. Concretely: the headline 5-bus result uses $N=500$ replications $\rightarrow$ 2000 scenarios $\rightarrow \approx 300$ test scenarios; Section VII’s confirmed IEEE 14-bus result uses the same $N=500$ replications (all 500 present) $\rightarrow$ 1993 of the possible 2000 scenarios (seven *physical-disturbance* scenarios, each from a different replication, were excluded because that scenario’s power flow did not converge – the other 3 scenarios in each of those 7 replications are unaffected and included) $\rightarrow$ 300 test scenarios for detection, 150 for localization. Wherever this paper states “ $n=X$ ” without qualification, X is the resulting test-scenario count, not the replication count – the two are only numerically equal here because both studies happened to use 500 replications.

IV. RESULTS I: THE TOPOLOGY-FUSION ADVANTAGE DOES NOT SURVIVE ITS OWN ABLATION

An initial near-perfect result was a methodology artifact, not a real effect. The first evaluation of topology_fusion reported 1.000 on every detection and localization metric, across three model families, with zero-width bootstrap confidence intervals, on a 72-scenario test set. We treated this as a red flag rather than a result: a defensive coding gap was found and fixed (a feature-exclusion filter for evaluation/label columns that was claimed in a comment but not enforced in code), but re-running showed it had never been the cause – the true cause, already anticipated in the pipeline’s own interpretation logic, was that the original attack scenarios had large, obvious state bias.

TABLE I
TOPOLOGY_FUSION VS. THE CORRECTED, TOPOLOGY-FREE ABLATION
(BEST MODEL PER CELL; 5-BUS, $n=300$ TEST)

Condition	Detection (bal. acc.)		Localization (top-1)	
	topo. fusion	res.+prior	topo. fusion	res.+prior
Hard	0.950	0.950	0.980	0.993
Harder ($+2\times$ noise)	0.953	0.953	0.980	0.993

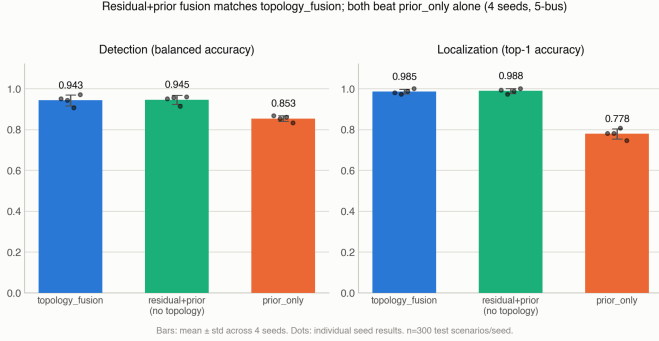


Fig. 1. Topology_fusion vs. the topology-free residual+prior ablation vs. prior_only alone, across 4 independent random seeds (5-bus, hard attack magnitude). Bars: mean $\pm$ std. Dots: individual seed results. $n=300$ test scenarios/seed.

Under a realistic attack magnitude, topology_fusion beat prior_only – a real, multi-seed-confirmed, but misleadingly-framed gap. Reducing attack magnitude roughly $3\times$, toward the sensor noise floor, and evaluating at $n=300$ test scenarios gave topology_fusion a clear lead over prior_only alone: detection 0.950 vs. 0.873 (non-overlapping 95% bootstrap confidence intervals), localization 0.980 vs. 0.780, both stable under a second, independent stress axis (doubled load-forecast uncertainty) and confirmed across four independent seeds (detection gap $+0.089 \pm 0.037$, localization gap $+0.207 \pm 0.032$, sign never flipping). All of this was true, and all of it was reported – but the comparison itself was incomplete: prior_only is missing residual information that topology_fusion has always had access to, so this gap could not, by construction, distinguish “topology helped” from “having residual information at all helped.”

The fair comparison is a near-exact tie. Table I compares topology_fusion against the corrected, genuinely topology-free residual_plus_prior ablation (Section III), for the same seed and both stress conditions.

Detection is an exact tie in both conditions; for localization, the topology-free ablation is *ahead* of topology_fusion, not behind it. For reference, both feature sets still comfortably beat prior_only alone (0.853–0.867 detection, 0.767–0.780 localization) and residual_only alone (0.687–0.690 detection, 0.700 localization) – the gain over either single baseline is real and large; it is simply not a topology effect.

The null result replicates across independent random seeds (Fig. 1). Repeating full data generation and evaluation for four independent seeds gives a topology_fusion-vs-residual_plus_prior detection gap of -0.003 ± 0.010

(range -0.013 to $+0.010$, crossing zero – the signature of noise, unlike the original, uncorrected topology_fusion-vs-prior_only gap, whose sign never flipped across the same four seeds) and a localization gap of -0.003 ± 0.007 (range -0.013 to 0.000 : never positive, so topology_fusion is at best tied, never ahead, in any of the 4 seeds). Both are practically null – 0.3 percentage points either way, an order of magnitude smaller than the original, uncorrected gaps ($+0.089$ detection, $+0.207$ localization) that motivated building this ablation in the first place – but they are not identical in character: detection is indistinguishable from zero in direction as well as magnitude, while localization has a small, direction-consistent lean against topology_fusion that is still too small to call a practically meaningful advantage for either side. Either way, this is the same distinction this project’s own $n=72 \rightarrow n=300$ correction (above) already established – a suspiciously clean-looking result does not survive a large-enough sample and a fair-enough control – now applied to a different kind of confound.

A structural explanation for the null result. Why would explicit neighbor-comparison features add nothing, given they are computed from genuinely different information – each bus’s residual/innovation relative to its neighbors, not just its own? On this fixed, small (5-bus, 4-line) topology, the answer is close to a linear-algebra fact rather than a coincidence. Predicting each of the 65 relational columns from residual_plus_prior alone, with 5-fold cross-validated ridge regression, gives a mean out-of-fold R^2 of 0.960 (median 1.000, minimum 0.652, 91% of columns above 0.8): a neighbor-mean or incident-edge aggregate over a fixed, small neighbor set is, on a network this size, close to an affine function of the same per-bus values residual_plus_prior already provides in full. A sufficiently flexible classifier (Hist-GradientBoosting, the best model throughout) can reconstruct nearly all of that relational information internally without ever being given it explicitly, which is exactly consistent with Table I’s exact tie. This explanation is specific to a fixed, small, low-redundancy topology, and does not by itself predict the independent IEEE 14-bus null result (Section VII), which uses an unrelated, size-agnostic feature design on a different, larger, meshed network – the two null results are consistent with each other, not mechanistically identical, and a network where relational features are *not* near-linearly redundant given local information remains an open, testable condition for when explicit topology-awareness might actually help.

V. RESULTS II: REAL-TIME FEASIBILITY – DIAGNOSING AND FIXING THE GAP

Following [4]’s finding that fast classifiers can still sit inside a slow end-to-end pipeline, we measured the full per-cycle decision pipeline (state estimation $\rightarrow$ feature computation $\rightarrow$ classification) against a one-cycle (20 ms at the network’s confirmed 50 Hz) protection-relevant computational target – not a claim that every protective relaying function requires exactly this latency, but the natural reference point for a per-cycle decision loop – with one-time network/model setup

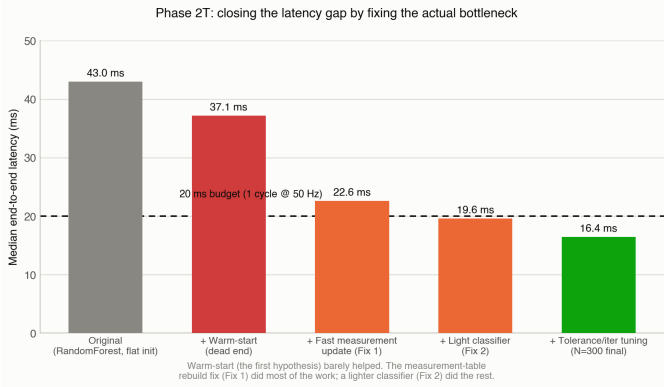


Fig. 2. Latency waterfall from the original pipeline to the final, tuned one. The first hypothesis (warm-start) barely helped; the measurement-table rebuild fix (Fix 1) did most of the work; a lighter classifier (Fix 2) did the rest.

excluded, matching how a deployed scheme would actually operate.

Initial measurement: median 42.98 ms, $2.15\times$ over budget, using the original pipeline (RandomForest classifier, tolerance= $1e-7$, maximum_iterations=40, flat state-estimator initialization every cycle).

The first hypothesis (slow convergence from a cold start) was wrong. Warm-starting the state estimator from the previous cycle’s converged solution produced only a $1.06\times$ speedup (37.14 ms), ruling out “far from the solution” as the dominant cost.

Profiling found an even split between two causes, only one of them algorithmic. Splitting the state-estimation call into its constituent steps showed the measurement-submission step cost 19.9 ms – essentially the same as the estimator’s own solve (19.4 ms). An ablation on maximum_iterations (40 $\rightarrow$ 5: no speed change; 40 $\rightarrow$ 3: convergence failure in 19/20 trials) showed the true iteration requirement is roughly 4–5 steps regardless of the 40-iteration cap, ruling out iteration count as the estimator-side cost driver.

The measurement-submission cost was pure, fixable software overhead. The measurement table was being rebuilt from scratch every cycle via a general-purpose element-creation API, called once per channel in a Python loop – even though the table’s structure is identical every cycle; only the values change. Replacing this with a one-time table construction followed by an in-place overwrite of the value column each cycle gave a $1158\times$ speedup on this sub-step (19.4 ms $\rightarrow$ 0.017 ms), verified to produce a bit-identical state-estimation result (max difference 0.0).

Combined with a lighter classifier (LogisticRegression, competitive with RandomForest per Section IV’s accuracy results and $14\times$ faster at inference), the pipeline reaches, at $N=300$ trials with 0 convergence failures: **16.41 ms median, 16.95 ms p95, 17.09 ms p99, 21.35 ms max** (Fig. 2) – median, p95, and p99 within the 20 ms budget; a single outlier (out of 300) still exceeds it, traced to the state estimator’s own internal solve-time variability. This benchmark uses its own, separately implemented 35-feature representation

TABLE II
RECALL ON A WITHHELD VS. TRAINED-ON ATTACK TYPE

Attack type	Features	Withheld	In training
Naive corruption	topo. fusion	0.000	0.827
Naive corruption	res.+prior	0.000	0.827
Stealth FDIA	topo. fusion	0.033	1.000
Stealth FDIA	res.+prior	0.047	1.000

(7 local-plus-neighbor statistics per bus $\times$ 5 buses) built for the timing measurement specifically, not `topology_fusion` or `residual_plus_prior` verbatim; feature computation already costs only 0.086 ms median, next to 16.14 ms for state estimation and 0.17 ms for inference, so the exact feature set used for classification is a second-order cost driver either way – the qualitative conclusion (state estimation, not the feature set or the classifier, is what the latency budget is spent on) does not depend on which of Section IV’s feature sets is deployed.

Positioning. [4] measures the same category of gap (sub-15 ms classification embedded in 50–90 ms pipelines) on an industrial-grade EMT simulator and recommends hardware acceleration. This paper’s finding – that roughly half of a comparable gap traced to an avoidable software inefficiency rather than any algorithmic or numerical cost – suggests pipeline profiling, not only hardware acceleration, deserves consideration first. A second, independent data point supports the “classifier inference itself is cheap” half of this argument: a knowledge-distilled LightGBM detector for virtualized microgrids reports 54–67 ms per 1000 samples (~ 0.06 ms/sample) for inference alone [22] – the same order of magnitude as this paper’s own model-inference sub-step (0.20 ms/sample), and nearly two orders of magnitude below this paper’s state-estimation sub-step (16 ms). Both results point the same way: once a classifier is lightened, the remaining latency budget is spent upstream of the model, not in it.

VI. RESULTS III: THE ADVANTAGE DOES NOT EXTRAPOLATE TO UNSEEN ATTACK TYPES

To test whether Section IV’s detector generalizes beyond the two attack families it was trained on (as opposed to merely discriminating them well), we trained HistGradientBoosting with one entire attack family withheld from training and validation, then measured recall on exactly that withheld family (Table II, Fig. 3). Run for both `topology_fusion` and the topology-free `residual_plus_prior` ablation, to check directly whether the collapse below is specific to one feature set or, as Section IV’s in-distribution finding would predict, behaviorally identical for both.

The detector recognizes the statistical signature of attack types represented in its training data – including the stealthy one, very well – but does not extrapolate to a mechanism it has not seen, and this holds identically whether or not topology-relational features are included: the two feature sets land within 1–1.4 percentage points of each other on every cell of Table II, confirming Section IV’s prediction that they are behaviorally interchangeable, here for a question –

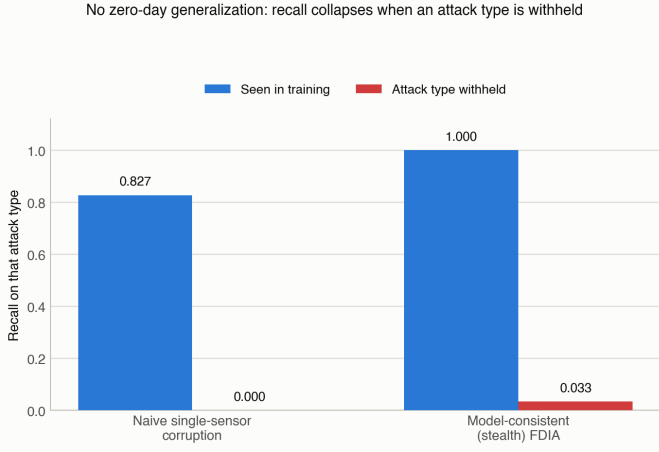


Fig. 3. Recall collapses when an attack type is withheld (topology_fusion shown; residual_plus_prior is behaviorally identical, Table II).

generalization to an unseen mechanism – that has nothing to do with topology specifically. We report this as a hard scope limit on any deployment claim built on Section IV’s results: within this project’s own two-family attack taxonomy, this is a complete absence of zero-day capability, not a partial one, for either feature set. This complements, from the opposite direction, a related generalization gap documented in the fault-classification literature: an XGBoost-based classifier trained and evaluated on the *same* distribution still drops from 96.8% (synthetic data) to 85.2% (physical data) [23] – i.e., even without an unseen attack *type*, a synthetic-to-real distribution shift alone costs over 11 points. Between that gap and this paper’s own 0–4.7% held-out-type result, generalization – of any kind – looks like the open problem for this class of detector, not an edge case. A deployment-aware zero-day evaluation in the adjacent EV/V2X charging-infrastructure domain [24] is a useful methodological reference for what a more rigorous version of Section VI’s test could look like.

VII. RESULTS IV: CROSS-NETWORK EVALUATION REVEALS TASK- AND NETWORK-SPECIFIC EFFECTS

Everything in Sections IV–VI uses the custom 5-bus microgrid. We separately tested whether explicit topology-relational features help at a larger, standard scale, escalating sample size across three passes until it matched Section IV’s own 300-scenario test-set size – using a completely independent, size-agnostic feature-engineering design, built before this section’s 5-bus correction and never adjusted to match it.

Test-system selection. The CIGRE European MV benchmark network was tried first and rejected for a concrete engineering reason: it represents internal switches via auxiliary buses (15 pandapower-level buses map to 18 internal power-flow-case buses), which breaks this project’s admittance-matrix-indexing code. IEEE 14-bus has no such mismatch and is, independently, the single most commonly used benchmark in the power-system state-estimation/FDIA literature specifically.

TABLE III
5-BUS VS. IEEE 14-BUS VS. IEEE 30-BUS: BEST RESULT PER TASK, AT THE SAME $n=300$ -TEST-SCENARIO SIZE THROUGHOUT (BAL. ACC. / TOP-1)

	topo.	res.+prior	prior	res.
Det., 5-bus	0.950	0.950	0.867	0.690
Det., IEEE-14 (4-seed mean)	0.741	0.750	0.707	0.750
Det., IEEE-30 (4-seed mean)	0.756	0.754	0.709	0.750
Loc., 5-bus	0.980	0.993	0.780	0.700
Loc., IEEE-14 (4-seed mean)	0.553	0.532	0.528	0.532
Loc., IEEE-30 (4-seed mean)	0.532	0.513	0.515	0.518

IEEE-14, confirmed across 4 independent seeds (Table IV): for detection, topology_fusion trails the best non-relational alternative each seed (gap -0.015 ± 0.004 , sign never flips); for localization, it leads each seed (gap $+0.018 \pm 0.010$, sign never flips either way) – both directions as seed-robust as the 5-bus result, just smaller in magnitude. IEEE-30, re-tested across the same 4 seeds (Table V) after an initial single run appeared to match IEEE-14’s pattern, does *not* replicate it: the detection gap changes sign twice (-0.007 to $+0.013$) and the localization gap changes sign once (-0.020 to $+0.020$) across the 4 seeds – neither is distinguishable from zero at this replication count, unlike IEEE-14’s uniformly-signed gaps.

IEEE-30’s rows above are left unbolded for this reason: the apparent column-mean leader is not a seed-robust winner.

Method. All WLS/measurement/attack machinery is network-size-agnostic by construction and was reused unmodified. The original fixed per-bus feature columns do not generalize past 5 buses and were replaced with a size-agnostic design: global summary statistics of the residual/innovation arrays, plus, for localization, a trained per-node classifier ranking every bus’s probability of being the attack target.

A second, independent instance of the same bug. This feature design is separately implemented from Section III’s regex-based column split, but auditing it after finding that bug surfaced an analogous flaw here too, in a cruder form: residual_only/prior_only were defined to include the single-argmax-node’s neighbor-comparison context (detection) and, for localization, every node’s neighbor-mean, neighbor-max, local-minus-neighbor, and degree columns directly (node_res_nb_mean, node_res_nb_max, node_res_local_minus_nb, node_n_neighbors, and their innovation-side counterparts) – an explicit list inclusion this time, not a name-substring collision, but the same structural problem: topology_fusion was, again, never compared against a condition holding all the same local information with no relational content. Fixed identically to Section III: relational/degree columns excluded from residual_only/prior_only by an explicit "neighbor" name marker (which also catches n_neighbors), their union added as a fourth residual_plus_prior set, topology_fusion left unchanged. All numbers below are from the corrected pipeline, re-run in full at $N=500$.

Result, escalating both sample size and seed count, same direction every time (Table III, Fig. 4; replication counts per the terminology defined in Section III). Two early passes – 80 replications (~ 24 test scenarios) and 200 replications (60–120 test scenarios) – predate the correction above and compared only the original (uncorrected) residual_only/prior_only/topology_fusion; both were

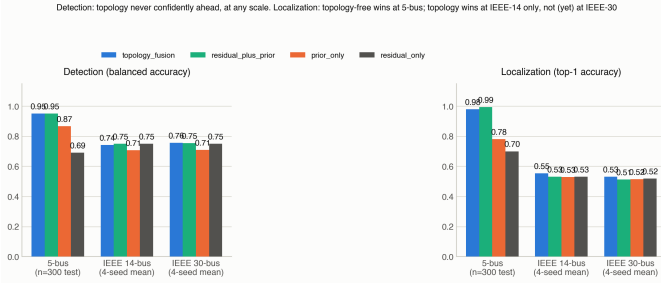


Fig. 4. 5-bus vs. IEEE 14-bus vs. IEEE 30-bus, all 4-seed means, all four feature sets. Detection: `topology_fusion` never confidently ahead, at any scale. Localization: the topology-free ablation is narrowly, consistently ahead at 5-bus (gap -0.003 ± 0.007); `topology_fusion` keeps a small, seed-robust lead at IEEE-14 only (Table IV) – IEEE-30’s own 4-seed replication (Table V) shows no stable lead either way, so the reversal is confirmed at one network, not a general property of scale.

TABLE IV
IEEE 14-BUS, `TOPOLOGY_FUSION` VS. THE BEST NON-RELATIONAL ALTERNATIVE EACH SEED, 4 INDEPENDENT SEEDS (500 REPL. EACH)

Seed	Detection gap	Localization gap
20260921	−0.013	+0.013
42	−0.013	+0.013
777	−0.011	+0.033
2024	−0.021	+0.013
mean ± std	−0.015 ± 0.004	+0.018 ± 0.010

inconclusive-to-suggestive purely on sample-size grounds, the same underpowered regime that had already produced, and been corrected out of, a false reversal in Section IV – a statement about N , not about the feature-definition fix that came later. A single corrected pass at 500 replications (300 test scenarios, matched to Section IV’s own) gave the single-seed numbers in Table III, with the localization margin narrow enough (0.560 vs. a best non-relational alternative of 0.547) that we did not initially trust it without the kind of seed replication Section IV used at 5-bus.

Re-running at 500 replications across the same 4 seeds used at 5-bus (Table IV) resolves that doubt in both directions: the detection gap is negative in all 4 seeds (`topology_fusion` *trailing* the best alternative, never leading) and the localization gap is positive in all 4 (`topology_fusion` leading, never trailing) – neither sign flips once. Detection is unambiguous, as it was in the single-seed pass: residual-only is competitive with or better than `topology_fusion` every time. Localization is the one place in this paper where `topology_fusion` shows a real, seed-robust edge over every non-relational alternative – small ($+0.018$ mean, versus 5-bus’s $+0.207$ -magnitude effect using the old, uncorrected comparison) but no longer dismissible as single-run noise.

Interpretation: two independently-built pipelines, corrected independently, agree on detection – and localization has a real, seed-confirmed reversal. Section IV (5-bus) and this section (IEEE 14-bus) were built with unrelated feature-engineering code – regex-derived fixed per-bus columns vs.

size-agnostic global summaries – and, caught only during this audit, each carried its own version of the same conceptual flaw, fixed separately above and in Section III. With both corrected, the two sections no longer read as “does topology help” and “does that transfer to scale” – they read as the same question, asked and answered twice with independent code, but the answer is task-dependent, not uniform. *Detection* agrees cleanly at both scales: `topology_fusion` carries no confidently-attributable advantage over a fair, non-relational baseline, at 5-bus (an exact tie) or at IEEE-14 (a consistent, 4-seed-robust deficit, Table IV). *Localization* does not agree – and, now that Table IV replaces the original single run, we are confident that disagreement is real rather than an artifact of under-replication: the topology-free ablation is narrowly but consistently ahead at 5-bus (gap -0.003 ± 0.007 , never favoring topology across 4 seeds, though small in absolute terms), while `topology_fusion` keeps a small but seed-robust lead at IEEE-14 (gap $+0.018 \pm 0.010$, sign never flipping across 4 seeds) – a real reversal in direction between the two networks, even though neither gap is large. That the same category of comparison mistake was made independently in two separately-written pipelines is itself informative – it suggests this is an easy trap when building a “topology-aware” feature set generally, not a one-off slip specific to one script – and the fact that correcting it produces a genuine, replicated task-by-scale reversal, rather than a uniform null result, is a more interesting and more credible outcome than either a clean “topology never helps” or a clean “topology always helps” would have been. This is consistent with, and sharpens, a 2025 field-wide scoping review’s [20] independent finding that cross-topology generalization is essentially untested across the ML-for-power-protection literature it surveyed – this study is a direct, sample-size-matched, cross-implementation attempt at a gap the field has already named, not a self-inflicted complication and not a preliminary pilot.

A third network, replicated: the single-run corroboration does not hold up. The same pipeline, re-run unmodified on IEEE 30-bus (`pandapower.networks.case30`, confirmed compatible with `h_ac()` the same way IEEE 14-bus was – `net.bus` count matches the internal power-flow-case bus count, so no CIGRE-MV-style indexing break), was first run at a single seed (20260921, this project’s own primary seed throughout) and appeared to reproduce both halves of the IEEE-14 result closely: detection gap -0.007 , localization gap $+0.020$. Given this project’s own repeated lesson about trusting single-run, small-margin results – the identical caution that motivated Table IV in the first place – we re-ran it across the same 4 seeds used at IEEE-14 (Table V) before accepting that corroboration.

It does not survive replication: the detection gap changes sign twice across the 4 seeds (mean $+0.001 \pm 0.009$) and the localization gap changes sign once (mean $+0.005 \pm 0.018$) – neither is a seed-robust effect by the same standard applied everywhere else in this paper. The original single run (seed 20260921) simply happened to land close to IEEE-14’s pattern by chance, in both directions, at once. We report this as a

TABLE V
IEEE 30-BUS, TOPOLOGY_FUSION VS. THE BEST NON-RELATIONAL
ALTERNATIVE EACH SEED, 4 INDEPENDENT SEEDS (500 REPL. EACH)

Seed	Detection gap	Localization gap
20260921	-0.007	+0.020
42	+0.000	+0.007
777	+0.013	+0.013
2024	-0.003	-0.020
mean $\pm$ std	+0.001 $\pm$ 0.009	+0.005 $\pm$ 0.018

genuine null result at IEEE-30, not a corroboration: the localization reversal found at IEEE-14 does not (yet) generalize to a second standard network, and whether that means the reversal is specific to IEEE-14, or that IEEE-30 needs more replications per seed to resolve an effect this small, is an open question – one this project’s own multi-seed discipline caught before it reached the paper’s headline claim, rather than after.

VIII. DISCUSSION

Read together, Sections IV–VII converge on a finding that is uniform for one task and genuinely not for the other, reached across three networks and two separately-written pipelines that each turned out to need the same category of fix. For *detection*: *explicit topology-relational features do not measurably improve cyber-physical event detection on this class of problem, once measured against a fair, same-information ablation* – not at a small 5-bus network (an exact tie), not at a larger, meshed IEEE 14-bus network built with an unrelated feature design (a consistent, 4-seed-robust deficit), and not at IEEE 30-bus either (a null result, gap $+0.001 \pm 0.009$, sign unstable). For *localization*, the honest answer is network-specific rather than a simple function of scale: the topology-free ablation is narrowly but consistently ahead at 5-bus (gap -0.003 ± 0.007), topology_fusion keeps a small but seed-confirmed lead at IEEE-14 (gap $+0.018 \pm 0.010$) – a real reversal, not noise – and IEEE 30-bus, larger than IEEE-14 by node count, shows no reliable direction either way (gap $+0.005 \pm 0.018$, sign unstable across seeds). If the mechanism were simply “bigger or more-meshed networks favor topology,” IEEE-30 should show at least as strong a reversal as IEEE-14, not a weaker and unstable one; it does not, which is itself evidence against a pure network-size story and for something more specific to IEEE-14 in particular. What helps detection at 5-bus, considerably, is combining residual and prior-innovation information: a real, multi-seed-confirmed ~ 0.09 gain over either alone (Section IV) – not a topology-specific effect, though also not a uniform one: at IEEE-14, the same fusion adds nothing measurable over residual_only alone (both mean 0.750). Section IV’s redundancy analysis gives the 5-bus null results a structural, not just empirical, footing: on a fixed, small topology, a relational aggregate is close to a linear function of information a flexible classifier already has, so there is little left for an explicit relational feature to contribute. The natural hypothesis – that this redundancy weakens on a bigger, more meshed network, explaining why

IEEE-14’s localization result comes out the other way – is directly testable with the same method, so we tested it rather than leaving it as speculation: repeating the identical out-of-fold check on IEEE-14’s own relational features (each node’s neighbor-mean/-max and local-minus-neighbor values, predicted from all 14 nodes’ local residual/innovation values) gives a mean R^2 of 0.959 (median 0.996, 96% of columns above 0.8) – statistically indistinguishable from the 5-bus figure (0.960), not lower. **The redundancy hypothesis does not explain the reversal; linear information content is not the mechanism.** A more likely explanation is discoverability rather than availability: HistGradientBoosting – a tree-based model with no built-in notion of “neighbor” – must find the right subset of a flat, unstructured input to combine for each target node. Whether that search gets reliably harder with more nodes is, on its own, only a partial explanation: IEEE-30’s own 30-node version of the same search does not show a *stronger* reversal than IEEE-14’s 14-node one, only a less stable one, which a pure “harder search at more nodes” story does not predict. Being handed the relational feature directly still removes that search entirely, so discoverability remains a plausible mechanism specifically at IEEE-14, but it is not, on this evidence, a clean function of network size either. This is a genuine open question this paper raises rather than resolves: the mechanism that explains detection’s null result at all three scales (Section III, Section IV) does not, on direct testing, explain localization’s IEEE-14-specific reversal, and what does – including why it does not extend to an even larger network – remains for future work.

The paper’s original framing – “does topology help, and where does that stop” – was itself built on an assumption (that residual_only/prior_only were topology-free) that turned out to be false by construction, not disproven by measurement; the more important finding here is not a scale boundary on a real effect, but the absence of the effect originally claimed, caught only by building the specific ablation a skeptical reviewer would have asked for regardless. We think this pattern – check the suspiciously good result before trusting it, then build the control condition the result was never actually tested against – is itself a methodological point worth stating explicitly, and a cheap one: the ablation that overturned this paper’s own headline claim required no new data collection, only a corrected column filter and one additional feature set trained on data already in hand.

IX. LIMITATIONS

- 1) **Scale:** Section VII’s IEEE 14-bus study independently cross-validates Section IV’s corrected 5-bus *detection* finding using an unrelated feature-engineering design (itself independently found to need, and given, the same correction – Section VII), now confirmed across 4 seeds (Table IV), not just one run. *Localization* does not cross-validate the same way – it shows a genuine, 4-seed-robust reversal instead (topology-free ahead narrowly at 5-bus, topology_fusion ahead narrowly at IEEE-14) – which we read as a real, network-specific effect,

not noise, though not one explained by scale alone (see Discussion). IEEE 30-bus, re-tested across the same 4 seeds after an initial single run appeared to corroborate IEEE-14, does not replicate that pattern: neither gap holds a consistent sign there (Table V). Three network topologies have now been tested to the same 4-seed standard; the localization reversal is confirmed at exactly one of them (IEEE-14), a null result at the other two (5-bus, IEEE-30); none uses a real (non-synthetic) network or load profile, and whether IEEE-30's null result reflects a genuinely IEEE-14-specific effect or simply needs more replications per seed to resolve is open.

- 2) **No zero-day generalization** (Section VI): detection/localization numbers hold only for attack types represented in training – verified for both `topology_fusion` and `residual_plus_prior`, which collapse identically (Table II); localization was not separately re-tested for zero-day generalization (only detection was), and the underlying attack taxonomy is still limited to two families (item 3 below).
- 3) **Single testbed family, two attack types**: restricting which measurements a stealth attack compromises is a physical consequence of the AC-consistent attack construction, not a tunable knob, and was not varied.
- 4) **Latency results** (Section V) characterize one Python/pandapower implementation on one machine, not a hardware or RTOS claim, and were not re-benchmarked on the smaller `residual_plus_prior` feature set (Section V argues this would not change the conclusion, since state estimation, not classification, dominates the budget, but this is an argument, not a re-measurement).
- 5) **Related-work verification is partial**: of ~ 57 sources surveyed during this project, 41 were read past search-summary level; two invented or misattributed figures, and one unverified qualitative literature-positioning claim (Section II), were caught and corrected during that process, which is grounds for treating any remaining unstarred source as provisional.

X. CONCLUSION

Across a small custom inverter-dominated microgrid and two standard, larger, meshed IEEE test systems (14-bus and 30-bus), explicit topology-relational features show no measurable detection advantage over a fair, same-information ablation (residual and prior-innovation features combined) – a finding reached independently at all three scales, with two unrelated feature-engineering implementations, and confirmed across four independent seeds at every scale tested. For localization, the honest result is a genuine, replicated reversal at exactly one of the three networks tested, not a general property of larger or standard networks, or of scale in general: the topology-free ablation is narrowly but consistently ahead on the small custom network, and `topology_fusion` keeps a small but seed-robust advantage at IEEE 14-bus – but an equally careful 4-seed replication at IEEE 30-bus, larger than IEEE-14 by node count and

which an earlier single run had seemed to corroborate, shows no consistent-sign effect in either direction. We take this as a caution against generalizing a two-scale pattern into a network-type rule from a single additional data point, and as a third instance, within this same project, of an unreplicated single-run result not surviving the replication standard the project's own earlier corrections established. Neither the uniform detection result nor the scale-specific localization one was the project's starting hypothesis: an initial, multi-seed-validated positive result (`topology_fusion` beating `prior_only` alone, at 5-bus, on both tasks) was real, but built on an incomplete comparison, exposed only once a genuinely topology-free control condition was constructed and a feature-labeling bug it revealed was corrected – independently, in two separate pipelines. Separately, the resulting decision pipeline can be brought within a one-cycle protection-relevant computational target once its actual bottleneck – a software inefficiency, not the model or the estimator's numerics – is found and fixed, and the detector does not extend to attack types unseen in training. We think the corrected, mostly-negative-but-not-uniformly-negative version of this result is more useful, and more publishable, than either an unqualified positive or an unqualified null result would have been: it identifies a specific methodological trap (comparing an information-rich feature set against baselines that were not actually information-poor) that is easy to fall into and cheap to check for, and it shows that trap does not always lead to the same answer once corrected – topology-awareness's value, on this evidence, is task- and scale-specific, not simply present or absent.

ACKNOWLEDGMENT

Experiment design, data generation, and evaluation for this work were carried out with AI-assisted software engineering and literature review (Claude, Anthropic); all reported quantitative claims trace to scripts and result files in the accompanying repository, and all literature claims marked as independently read in `RELATED_WORK.md` were verified against primary sources rather than taken from search summaries alone. The feature-labeling bug central to this paper's corrected finding (Section III) was caught during an AI-assisted code review while implementing the `residual_plus_prior` ablation requested as a next step in this project's own research process; the correction and re-analysis in Sections IV and VII were completed, and the original positive framing withdrawn, before this manuscript was finalized.

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